Amey Mishra

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Education

HARMONY SCHOOL OF ADVANCEMENT | Houston, TX | Expected May 2027

Valedictorian | GPA: 4.75 | SAT: 1520 | Mechanical/Aerospace Engineering

Advanced Coursework — Multivariable Calculus, Linear Algebra, College Physics II, AP Calculus BC, AP Research, AP Cybersecurity, AP Environmental Science, AP English Literature.

Technical Skills

Mechanical / CAD — Fusion 360, Onshape, Revit, mechanism design, DFM, 3D printing, prototyping, actuator integration.

Programming / Controls — Python, Java, Arduino, OpenCV, sensor feedback, autonomous navigation, path planning, inverse kinematics.

High-Impact Experience

FOUNDER, CAPTAIN & MECHANICAL LEAD | 2023 - Present

FIRST Tech Challenge Team 25679 The Cosmobots | thecosmobots.com

- Led robot architecture, mechanical design, fabrication, testing, strategy, and team operations; top worldwide performance among 8,000+ teams, World Championship 4th-ranked alliance, Carolinas Premier Champion, and FTC Hall of Fame.
- FIRST Leadership Award Finalist; reduced a high-cycle transfer mechanism from ~0.75 s to ~0.30 s through iterative CAD, actuator- placement studies, and field testing.

FOUNDER & CEO | 2024 - Present

Cosmobots Robotics | cosmobotsrobotics.com

- Built a STEM nonprofit supporting four FTC teams and three rocketry teams; secured $50K+ in support, coordinated mentors/budgets/equipment, and built partnerships with Unitree Robotics, Stokes Education, Harmony Public Schools, Meiborg, and others.
- Generated 4,500+ organizational service hours and 30+ outreach events; reached 25,000+ people directly and students across seven countries.

LEAD RESEARCHER & SYSTEMS ENGINEER | 2025 - Present

Space Debris Research / NASA Space Apps - ORCA | orcadebriscleanup.space

- Published research on orbital-debris growth/collision risk; led ORCA, a capture-and-recycling concept using debris-analysis neural networks, orbital mechanics, and CAD/simulation of collection/recycling systems.
- NASA Space Apps Regional Winner and Global Semifinalist; developed mission feasibility, operations, implementation, and business- planning components.

ROBOTICS & AI RESEARCHER | 2026 - Present

Mentored by a Stanford PhD Student

- Developed a neural-network-powered humanoid arm with custom CAD/3D-printed structure, motors/sensors, adaptive inverse kinematics, Java/Arduino controls, and OpenCV perception.

DESIGN & MANUFACTURING INTERN | 2026 - Present

Pike Robotics

- Designed pivoting camera mounts and umbilical-management mechanisms in Fusion 360/Onshape for an industrial inspection robot operating inside 100+ ft tanks.

Additional Engineering & Leadership

Rice University Research — Electromagnetic energy-harvesting study; microcontroller test systems, simulation, conference presentation, and research manuscript submission.

RECF Aerial Drone Competition — Founder/Captain/Pilot; Python autonomous-flight library; 3 Regional Championships, 3 Excellence/All- Around Awards, 3 Nationals qualifications, Flight Operations Award, Judges Award.

Rocketry / TVC — American Rocketry Challenge team lead; designed/tested custom thrust-vector-control system; autonomous flight programs achieved ~97% successful execution in testing.

Science Olympiad — President and engineering competitor; 7 Regional Golds, 1 State Gold, 2 State Bronzes in build/engineering events.

Honors, Projects & Certifications

Honors — FIRST Leadership Award Finalist; FTC Hall of Fame / Carolinas Premier Champion; NASA Space Apps Global Semifinalist; Gene Kranz Scholarship; International Math Challenge Gold.

Projects — Autonomous mapping UAV; RC aerodynamics test aircraft; humanoid robot; NASA exoplanet CNN.

Certifications — Autodesk Fusion 360 & Revit; CompTIA Network+; OpenEDG & Microsoft Python; FAA TRUST; Adobe Photoshop/Illustrator; W3Schools JavaScript.